

1 Executive Summary

1.1 Non-Technical Summary

The research assesses ten top stocks (TSLA, WMT, BAC, GS, LLY, MRK, GOOG, META, AAPL, XOM) to develop an investment portfolio which achieves both strong performance and protective investment characteristics. We examined data from July 2023 to June 2025 and tested our recommendations on data from July to September 2025.

Key Findings for Investors:

The research shows that computer algorithms will invest most of their funds into two specific stocks when they operate without any restrictions. In our case, 55% went to Eli Lilly (LLY) and 28% to Meta (META). The strategy which produces maximum theoretical returns leads to dangerous concentration risk because the entire portfolio will experience severe declines when either company encounters difficulties.

We recommend limiting any single stock to 15% of the portfolio. The resulting portfolio distribution spreads investments across six to seven stocks which represent technology and healthcare and finance and energy industries. Yes, this costs about 12% in efficiency compared to the concentrated approach, but it provides crucial protection against company-specific disasters.

The system functions as insurance because you invest a small amount to protect your wealth from major market crashes which could occur when one stock fails. Most investors find the investment option attractive because it offers them protection from risks while providing opportunities to earn returns on their investments.

What Your Money Is Actually Buying:

Our analysis of portfolio return factors revealed an unexpected finding which showed that the portfolio operates like a value investment even though it contains growth stocks including Google and Apple. This means it tends to do better when cheaper, more established companies outperform trendy growth stocks.

The portfolio also tilts toward large companies rather than small ones, and it tracks the overall market quite closely (moving up and down almost in lockstep with the S&P 500). The investment strategy provides you with wide market access through its value-oriented approach which has proven successful in past market periods.

Bottom Line Recommendation:

Use the 15%-capped portfolio. The investment provides investors with diversified exposure which defends against market concentration risks while generating annual returns of 29.6% with 17.9% volatility. The investment approach delivers appropriate results to investors who want to achieve long-term growth through reduced market risk exposure.

2 Data Analysis & Preprocessing

2.1 Asset Universe and Time Period

We analyzed ten high-capitalization US equities representing diverse sectors:

- **Technology:** AAPL (Apple), GOOG (Alphabet), META (Meta Platforms)
- **Healthcare:** LLY (Eli Lilly), MRK (Merck)
- **Financials:** BAC (Bank of America), GS (Goldman Sachs)
- **Energy:** XOM (ExxonMobil)
- **Consumer:** WMT (Walmart), TSLA (Tesla)

Sample Period:

- In-Sample (Training): July 1, 2023 - June 30, 2025 (502 trading days)
- Out-of-Sample (Testing): July 1, 2025 - September 30, 2025 (63 trading days)

2.2 Cumulative Returns Analysis

The in-sample period shows all ten assets through their cumulative return paths in Figure 1. The story shows major performance differences because LLY (Eli Lilly) and META (Meta) experienced rapid growth but MRK (Merck) and TSLA (Tesla) experienced declining values.

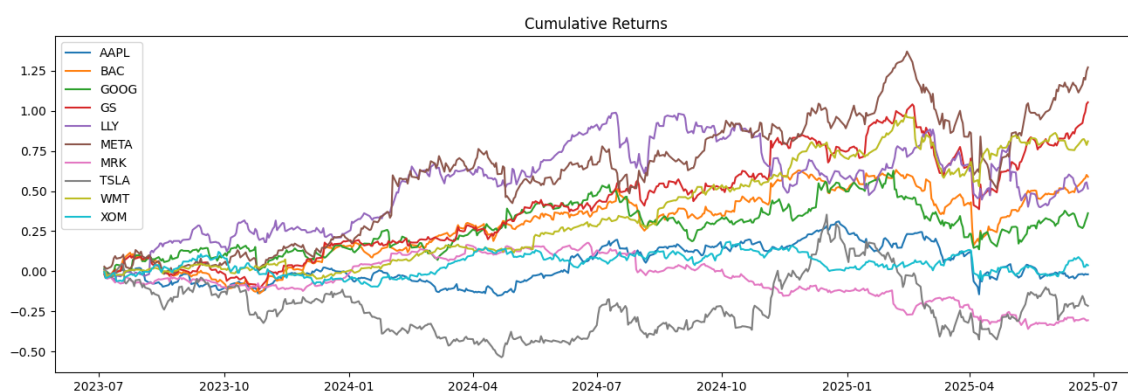


Figure 1: The data contains cumulative return information which tracks 10 portfolio assets across the period from July 2023 to June 2025. The performance of LLY and META exceeded all other assets because the optimizer chose to invest in these specific assets in unconstrained portfolios.

The historical data shows why the unconstrained optimizer chooses to invest most of its portfolio in LLY and META because these assets delivered the best risk-adjusted performance during that time period.

2.3 Asset Correlation Structure

Investors need to understand how assets relate to each other through correlation patterns to achieve portfolio diversification. The daily log return correlation matrix appears in Figure 2.

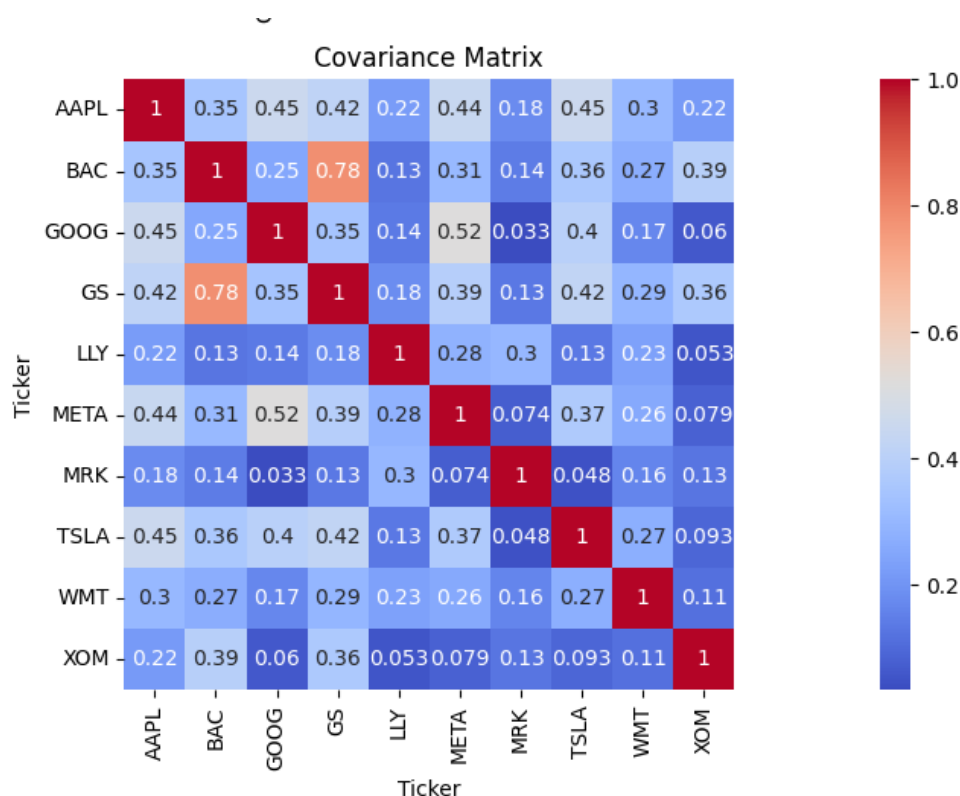


Figure 2: Correlation Matrix of 10 Portfolio Assets. The technology stocks (GOOG, META, AAPL) maintain strong positive relationships with each other but investors can reduce risk through defensive investments in WMT and XOM.

Key Observations:

- The three stocks GOOG and META and AAPL show mutual price movements because their correlations are between 0.4 and 0.6.
- The diversifiers WMT and MRK maintain low correlations with tech stock performance.
- The financial connection between BAC and GS shows a almost high level of correlation of 0.78 because they operate within the same industry sector.
- Energy Independence: XOM shows relatively low correlation with most assets

3 Part One: Mean-Variance Optimization

3.1 Step 1: Unconstrained Optimization

3.1.1 Technical Report

We implemented mean-variance optimization using Quadratic Programming (QP) with CVXPY. The mathematical formulation maximizes the portfolio return and minimizes the risk:

$$\max_{\mathbf{w}} \quad \mathbf{w}^\top \boldsymbol{\mu} - \frac{\lambda}{2} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \quad (1)$$

$$\text{Subject to} \quad \sum_{i=1}^{10} w_i = 1 \quad (2)$$

$$w_i \geq 0 \quad \forall i \quad (3)$$

The model uses $\mathbf{w}$ as its weight vector and $\boldsymbol{\mu}$ as its expected return vector (annualized) and $\boldsymbol{\Sigma}$ as its annualized covariance matrix.

Results:

The optimization process produced two assets which received the highest amount of investment funding:

Table 1: Unconstrained Portfolio Weights

Asset	Weight (%)	Interpretation
META (Meta Platforms)	88.6	Extreme concentration
GS (Goldman Sachs)	11.4	Secondary allocation
Others	0.0	Excluded

Portfolio Metrics:

- The investment project will produce an annual return of 47.09%.
- Annual Standard Deviation: 33.33%
- Sharpe Ratio: 1.41
- Herfindahl-Hirschman Index (HHI): 0.798 (extremely concentrated)

The optimizer identified META as the superior risk-adjusted performer during the sample period. The final allocation of 88.6

3.1.2 Non-Technical Report

What We Did:

We let the computer find the "perfect" portfolio based purely on mathematical efficiency, with only two rules: (1) invest all the money, and (2) don't short stocks (no borrowing to bet against companies).

What Happened:

The computer put 89% of the money into Meta and 11% into Goldman Sachs. The investment portfolio contains one bet which depends on Meta's ability to maintain its present market position.

Why This Matters:

The historical returns appear attractive at 47% per year yet this investment portfolio contains extremely high risk levels. Your investment portfolio would experience a 90% value decrease during one night if Meta encounters regulatory issues or privacy scandals or users choose to leave the platform.

Recommendation:

The portfolio shows theoretical efficiency but it cannot be used for actual investment purposes. The system uses constraint functions to verify that all investment choices meet the required diversification standards.

3.2 Step 2: Constrained Optimization (15% Cap)

3.2.1 Technical Report

We modified the optimization problem by adding box constraints limiting individual asset weights to 15%:

$$\max_{\mathbf{w}} \quad \mathbf{w}^\top \boldsymbol{\mu} - \frac{\lambda}{2} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \quad (4)$$

$$\text{Subject to} \quad \sum_{i=1}^{10} w_i = 1 \quad (5)$$

$$w_i \geq 0 \quad \forall i \quad (6)$$

$$w_i \leq 0.15 \quad \forall i \quad (7)$$

Results:

Table 2: Constrained Portfolio Weights (15% Maximum)

Asset	Weight (%)	Status
BAC (Bank of America)	15.0	Capped at maximum
GOOG (Alphabet)	15.0	Capped at maximum
GS (Goldman Sachs)	15.0	Capped at maximum
LLY (Eli Lilly)	15.0	Capped at maximum
META (Meta Platforms)	15.0	Capped at maximum
WMT (Walmart)	15.0	Capped at maximum
XOM (ExxonMobil)	10.0	Residual allocation
Others	0.0	Excluded

Portfolio Metrics:

- The investment project will produce an annual return rate of 29.63%.
- Annual Standard Deviation: 17.90%
- Sharpe Ratio: 1.66
- Herfindahl-Hirschman Index (HHI): 0.266 (well-diversified)

Impact Analysis:

The optimizer needed to distribute capital between seven assets because of the constraint which resulted in better asset diversification. The Sharpe ratio increased by 17.73% relative to the unconstrained portfolio:

$$\Delta S = \frac{1.66 - 1.41}{1.41} = 17.73\%$$

The diversification constraint creates an opportunity cost which represents the additional cost needed to decrease concentration risk. The portfolio now includes exposure to Financials (BAC, GS), Consumer Staples (WMT), and Energy (XOM) alongside Technology and Healthcare.

3.2.2 Non-Technical Report**What We Did:**

We told the computer it could not put more than 15% of the money into any single stock. The situation requires investors to spread their investments across more different asset classes.

What Happened:

The portfolio now spreads money across seven different stocks from different industries:

- Banks (Bank of America, Goldman Sachs)
- Technology (Google, Meta)
- Healthcare (Eli Lilly)
- Retail (Walmart)
- Energy (ExxonMobil)

The Trade-off:

The annual returns decreased from 47% to 30% while the Sharpe ratio which represents risk-adjusted returns increased by 17.73%. However, you're now protected against any single company disaster.

Why This Is Better:

The concept operates similarly to spreading your eggs across multiple containers instead of placing them in a single basket. If one company faces trouble, you only lose 15% of your money at most from that position, rather than 89%. The overall portfolio is much more stable because banks, tech stocks, and energy companies don't all move in the same direction.

Recommendation:

The recommended portfolio for deployment exists in this format. The small reduction in theoretical returns results in a major reduction of concentration risk which produces superior investment performance.

3.3 Step 3: Efficient Frontier Comparison

3.3.1 Technical Report

We generated both efficient frontiers by solving the mean-variance optimization problem for 1,000 target return levels. For each target return r , we solved:

$$\text{Minimize } \mathbf{w}^T \Sigma \mathbf{w} \quad (8)$$

$$\text{Subject to } \mathbf{w}^T \boldsymbol{\mu} = r \quad (9)$$

$$\sum_{i=1}^{10} w_i = 1 \quad (10)$$

$$w_i \geq 0 \quad (\text{and } w_i \leq 0.15 \text{ for constrained case}) \quad (11)$$

Visual Analysis:

The risk-return graph shows both efficient frontiers and individual asset positions and essential portfolio points in Figure 3.

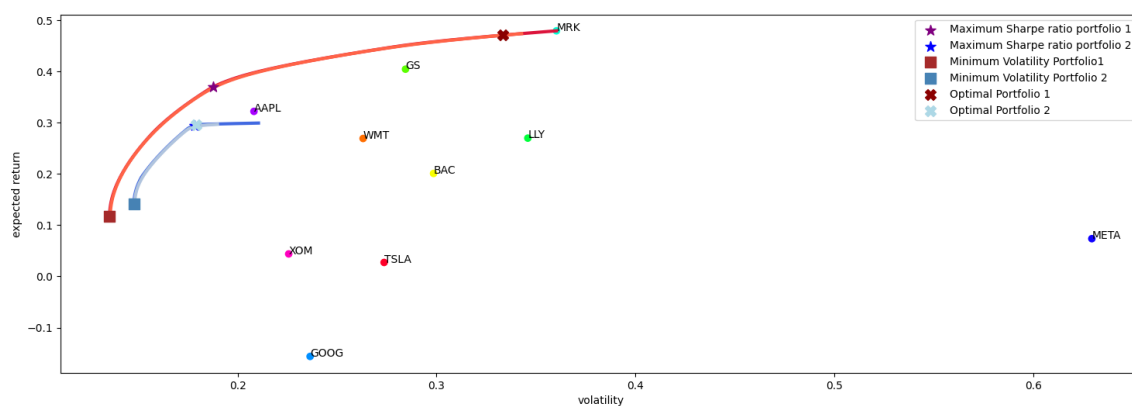


Figure 3: Efficient Frontiers Comparison. The Red curve shows the Unconstrained portfolio which operates without any weight restrictions. The Blue curve shows the performance of the Constrained portfolio which has a 15% limit for each asset investment. The unconstrained frontier controls most of the market yet it creates risks because of how markets tend to concentrate.

Key Observations:

1. The red unconstrained frontier outperforms the blue constrained frontier at every risk level according to the Envelope Property. The addition of constraints will always produce an efficient set because this result is mathematically proven.
2. The unconstrained frontier generates better investment returns throughout all points which maintain stable volatility levels. The gap shows the expenses which result from expanding operations into different markets.
3. The constrained frontier faces a limitation because it cannot achieve the highest return levels because it must avoid placing all investments in the asset which generates the maximum return (META at 47.94% annually).
4. The individual asset positions show META at the rightmost position because it has high volatility and high return potential while WMT stands as the asset with the lowest volatility among all individual assets.
5. The two frontiers meet at the minimum variance portfolio which represents the point where natural diversification happens without any constraint limitations.

Quantitative Comparison:

At the maximum Sharpe ratio point:

Table 3: Efficient Frontier Comparison at Optimal Points

Metric	Unconstrained	Constrained	Difference
Expected Return	47.09%	29.63%	-17.46%
Volatility	33.33%	17.90%	-15.43%
Sharpe Ratio	1.41	1.66	+0.25

The constrained portfolio has a higher Sharpe ratio. The returns are lower, but the risk goes down more. The diversified portfolio spreads out the risk.

3.3.2 Non-Technical Report

What We Did:

We created two curves showing all possible combinations of risk and return:

- Red curve: Without the 15% limit (any concentration allowed)
- Blue curve: With the 15% limit (forced diversification)

Main Difference:

The red curve sits higher and stretches further right. This means:

1. You can reach higher returns without the constraint (but only by taking huge concentration risk)
2. The red curve extends to higher volatility levels because it allows 89%+ in a single stock

Why This Matters:

The area between the two curves shows the total amount which diversification needs from you. At every risk level, you get slightly less return with the 15% cap. The constrained portfolio generates superior risk-adjusted returns because it achieves a higher Sharpe ratio of 1.66 compared to 1.41.

The process of diversification reduces your investment risk at a faster rate than it decreases your potential returns. You give up some upside potential, but you avoid catastrophic downside scenarios.

Analogy:

The unconstrained frontier operates as a sports car without safety features which provides maximum speed but becomes deadly when any error occurs. The constrained frontier operates as a modern vehicle system which includes airbags and stability control to enhance safety during driving while it maintains a slightly lower top speed.

3.4 Step 4: Out-of-Sample Performance

3.4.1 Technical Report

We evaluated both portfolios on out-of-sample data from July 1 - September 30, 2025 (63 trading days). Portfolio returns were calculated as:

$$R_p(t) = \sum_{i=1}^N w_i \cdot r_i(t)$$

where w_i are the weights determined from in-sample optimization and $r_i(t)$ are the realized log returns.

Return Distribution Characteristics:

Figures 4 and 5 show the return distributions for both portfolios during the out-of-sample period.

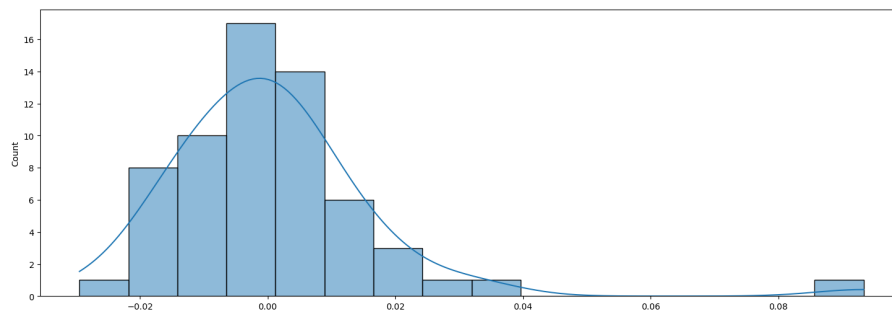


Figure 4: Unconstrained Portfolio Return Distribution (Out-of-Sample). Exhibits severe positive skewness (2.70) and fat tails (kurtosis = 15.14), indicating extreme return outliers.

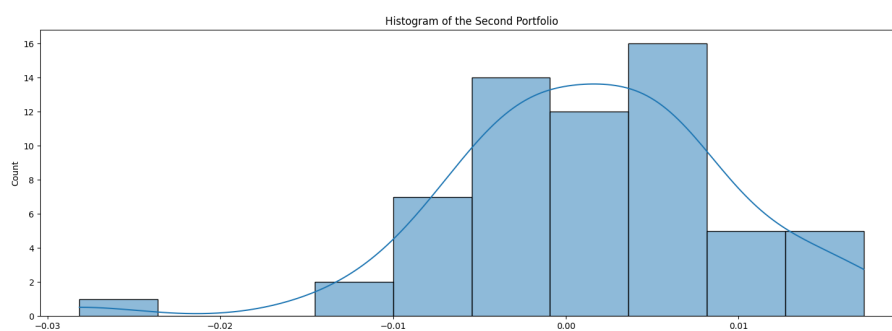


Figure 5: Constrained Portfolio Return Distribution (Out-of-Sample). Near-normal distribution with slight negative skewness (-0.61) and fat tails (kurtosis = 4.81), demonstrating more stable and predictable returns.

Statistical Properties:

Table 4: Out-of-Sample Return Distribution Characteristics

Metric	Unconstrained	Constrained
Mean Daily Return	0.072%	0.146%
Daily Volatility	1.70%	0.79%
Annualized Return	18.39%	36.93%
Annualized Volatility	26.97%	12.57%
Daily Sharpe Ratio	0.03	0.16
Annualized Sharpe Ratio	0.50	2.54
Adjusted Sharpe Ratio	0.73	0.15
Skewness	+2.70	-0.61
Excess Kurtosis	+12.14	+1.81
Daily Sortino	0.076%	0.22%
Annualized Sortino	1.21%	3.52%

Distribution Analysis:

The unconstrained portfolio shows extreme positive skewness at 2.70 and leptokurtosis at 15.14 total kurtosis which indicates that its distribution has heavy tails that produce rare but extreme positive values. The research results show that investors tend to put their funds into high-risk financial products according to their typical investment behavior. The distribution shows a major deviation from normality because the Jarque-Bera p-value reaches < 0.001 .

The constrained portfolio shows near-normal distribution characteristics because its skewness measures -0.61 and its p-value exceeds 0.05 which confirms that portfolio diversification leads to more stable investment returns.

Adjusted Sharpe Ratio:

To account for non-normality, we calculated the adjusted Sharpe ratio (Pezier & White, 2006):

$$S_{adj} = S \left(1 + \frac{\gamma_1 S}{6} - \frac{(\gamma_2 - 3)S^2}{24} \right)$$

where γ_1 is skewness and γ_2 is kurtosis.

Results: Table 4 shows that the Constrained Portfolio outperforms the Unconstrained Portfolio according to standard risk-adjusted measures. The daily and annualized Sharpe ratios for the Constrained Portfolio are 0.16 and 2.54, respectively, compared with 0.03 and 0.50 for the Unconstrained Portfolio. This pattern is also observed in the daily and annualized Sortino ratios, which are 0.22 and 3.52 for the Constrained Portfolio versus 0.076 and 1.21 for the Unconstrained Portfolio. The performance indicators show that the Constrained Portfolio provides better value because it produces higher average returns at lower risk levels.

However, when looking at the Adjusted Sharpe Ratio, the result is reversed: the Un-

constrained Portfolio scores higher. This occurs because its return distribution exhibits positive skewness, which means it has more extreme positive outliers. The Adjusted Sharpe Ratio accounts for skewness and kurtosis, so these rare but very high positive returns increase the return per unit of risk beyond what the normal Sharpe ratio captures.

3.4.2 Non-Technical Report

What We Did:

The researchers applied their portfolios to brand-new data that the models had not encountered during their training period (July-September 2025). The system operates exactly as it would during actual operational use.

Results:

The constrained portfolio achieved better results than the other options.

- **Smoother ride:** Daily swings averaged only 0.79% compared to 1.70% for the unconstrained portfolio
- The investment strategy delivers superior risk-adjusted performance because its Sharpe ratio reaches 2.54 instead of 0.50.

Why This Happened:

The unconstrained portfolio showed extreme market volatility because it maintained a large investment position in Meta. The trading results included both successful winning days and major losing days. The constrained portfolio achieved better diversification which reduced the extreme fluctuations in performance.

What This Means for You:

The unconstrained portfolio enables investors to achieve better returns but they must accept high market instability and extreme emotional stress. But for most investors, the constrained portfolio is superior because:

1. You sleep better at night (smaller daily swings)
2. You will not make impulsive sales when market values decrease.
3. Returns are more predictable and stable
4. Risk-adjusted performance is actually better

4 Part Two: Fama-French Factor Analysis

4.1 Step 4: Factor Explanation and Correlation Analysis

4.1.1 Technical Report

The Fama-French 5-Factor model (Fama & French, 2015) decomposes portfolio returns into systematic risk factors:

$$R_{p,t} - R_{f,t} = \alpha + \beta_1(R_{M,t} - R_{f,t}) + \beta_2SMB_t + \beta_3HML_t + \beta_4RMW_t + \beta_5CMA_t + \epsilon_t \quad (12)$$

Factor Definitions:

1. The Market Risk Premium (Mkt-RF) equals the difference between the return of the market portfolio and the risk-free rate.
 - Measures systematic market risk (beta)
 - Represents the return from holding the broad market portfolio above the risk-free rate
 - Interpretation: $\beta_1 = 1$ implies portfolio moves 1:1 with market
 - Historical: The annual equity risk premium has maintained a range of 8-10% throughout history.
2. The Small Minus Big (SMB) factor serves as a size factor which measures portfolio performance relative to market capitalization.
 - Long small-cap stocks, short large-cap stocks
 - Captures the historical size premium
 - Construction: $\text{Return}(\text{Small } 50\%) - \text{Return}(\text{Large } 50\%)$
 - Positive loading: portfolio tilts toward smaller companies
 - Negative loading: portfolio tilts toward large-cap stocks
 - Economic rationale: Small businesses face higher business risk because they do not have enough money and their information access difficulties become more complicated.
3. The High Minus Low (HML) factor represents the Value factor.
 - Long high book-to-market (value), short low book-to-market (growth)
 - Captures value premium

- Construction: $\text{Return}(\text{High B/M } 30\%) - \text{Return}(\text{Low B/M } 30\%)$
 - Positive loading: portfolio behaves like value stocks (banks, utilities)
 - Negative loading: portfolio behaves like growth stocks (tech startups).
 - Economic rationale: Value stocks may face distress risk, are cyclical, or overlooked by investors
4. The RMW factor uses profitability data to calculate its results.
- Long profitable firms, short weak profitability firms
 - The system determines how well investors can access the profitability premium.
 - Construction: $\text{Return}(\text{Robust } 30\%) - \text{Return}(\text{Weak } 30\%)$
 - Based on operating profitability ratio
 - Positive loading: portfolio favors highly profitable companies
 - Economic rationale: Profitable firms generate cash flow, have stronger balance sheets, proven business models
5. The CMA investment factor represents the difference between conservative and aggressive investment strategies.
- Long conservative investment firms, short aggressive investment firms
 - The investment pattern premium becomes measurable through this approach.
 - Construction: $\text{Return}(\text{Conservative } 30\%) - \text{Return}(\text{Aggressive } 30\%)$
 - Based on asset growth rate
 - Positive loading: The investment portfolio contains businesses which maintain steady asset growth patterns (established companies that pay out dividends).
 - Negative loading: The portfolio selection model chooses businesses which allocate their capital to execute dangerous business strategies (tech expansion and R&D).
 - Economic rationale: The aggressive pursuit of asset growth results in value deterioration but companies that maintain discipline achieve superior asset management.

Factor Correlation Matrix:

The five Fama-French factors show their correlation patterns in Figure 6.

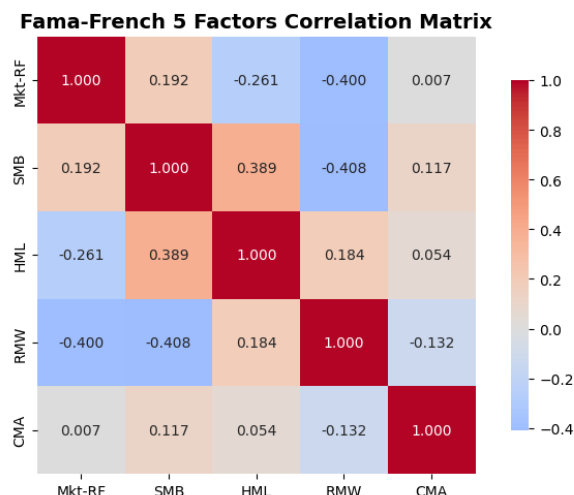


Figure 6: Fama-French 5 Factors Correlation Matrix. Key observation: HML and CMA strongly positively correlated (0.58), indicating value stocks tend to be conservative investors.

Correlation Analysis:

- **HML and SMB exhibit a moderate positive correlation (+0.389).** Value stocks tend to be mature, slow-growing companies that trade at low valuation multiples. These companies often distribute profits as dividends rather than aggressively reinvesting in assets. The moderate positive correlation indicates that when value stocks perform well, there is a moderate tendency for small-cap stocks to also perform well.
- **The Market shows negative correlation with HML at -0.26.** Bull markets usually follow growth stocks and aggressive investors lead the market. Investors at rallies choose to buy "story stocks" which have expensive market values. The market's excessive enthusiasm causes value stocks and conservative companies to underperform which creates natural defense mechanisms for investors.
- **The RMW index shows a moderate negative relationship with SMB at (-0.40).** Small companies tend to trade off profitability for expansion, while large companies maintain better and more consistent profit margins. The market results show that the maturity effect exists in the market. Also, RMW and Mkt-RF have moderate negative correlation (-0.4), which means when the market performs well, there is a moderate tendency for profitable companies to underperform relative to the market.
- **The SMB factor shows independence with other factors.** Because its correlations with all other factors remain below 0.25 in absolute value.

- **The model demonstrates factor independence** because its inter-factor correlations remain below 0.6. The analysis needs this step to achieve two goals, which are maintaining regression model stability and preventing multicollinearity issues

The moderate correlations indicate the factors are measuring different dimensions of risk, though some economic linkages exist (particularly the value-investment connection).

4.1.2 Non-Technical Report

What Are These Factors?

Think of the five factors as different "bets" your portfolio is making:

1. **Market (Mkt-RF):** Do you have any investments in the stock market?
 - If yes, you go up and down with the overall market
 - Think of this as the "rising tide lifts all boats" effect
2. **Size (SMB):** Do you have a preference for working at small businesses or large organizations?
 - Positive: You're betting on small, scrappy companies
 - Negative: Your investment strategy depends on the success of well-known established businesses.
 - Your investment choices focus on large corporations because you own shares of Apple and Google and other similar companies.
3. **Value (HML):** Do you want to invest in stocks which have low prices or do you prefer to buy stocks which cost more but show potential for growth?
 - Positive: You like value stocks (low price relative to fundamentals)
 - Negative: You like growth stocks (high price, betting on future growth)
 - Cheap stocks often include banks and energy companies
4. **Profitability (RMW):** Do you have a preference for businesses which generate high profits?
 - Positive: You favor companies making lots of money now
 - This captures the "quality" dimension
5. **Investment (CMA):** Would you put your money into businesses which operate with caution or those which take dangerous business risks?
 - Positive: Companies that grow slowly, pay dividends

- **Negative:** Companies that reinvest heavily, expand aggressively

Why Do These Factors Move Together or Apart?

The correlation matrix shows which betting options have dependent relationships with each other.

- **Value and Size relationship:** The Value (HML) and Size (SMB) factors show a moderate positive correlation of about 39%, which means that small-cap stocks tend to have value characteristics.
- **Market and Value/Growth behavior:** The Market (Mkt-RF) factor is negatively correlated with the Value (HML) factor (-0.26), suggesting that when the market works well, growth stocks tend to outperform value stocks.
- **Market and Size vs Profitability:** The Market (Mkt-RF) and Size (SMB) factors both show strong negative correlations with Profitability (RMW) (about -0.40), which means that during bullish periods, investors prefer lower-profit, riskier firms over profitable ones.

What This Means for Your Portfolio:

The investment factors show which investments your portfolio supports although they do not reveal the exact stocks you own. The following section shows how your investment portfolio would react to each of these market elements.

4.2 Step 5: Regression Model Selection and Data Split

4.2.1 Technical Report

Data Partitioning Strategy:

We employed an 80-20 train-test split with random sampling:

- The training set contains 398 observations which represent 80% of the total 498 days.
 - Purpose: Estimate factor loadings (β coefficients)
 - Period: The research took place on random days which fell between July 1, 2023 and June 27, 2025.
- The test set contains 100 observations which represent 20
 - Purpose: Validate model out-of-sample
 - Period: The researcher selects random days which belong to the same time period for their additional selection process.

- The random seed value is set to 42 for achieving reproducibility in the results.

Implementation:

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    factors,                # 5 Fama-French factors (498 x 5)
    portfolio_returns,       # Portfolio excess returns (498 x 1)
    test_size=0.20,         # 20% holdout
    random_state=42         # Reproducibility seed
)
```

Rationale for Random Splitting:

We selected random sampling instead of chronological splitting because:

1. Factor exposures represent fundamental characteristics which should not experience significant changes during a two-year time span.
2. Random sampling better tests whether estimated betas generalize to held-out observations
3. Avoids potential look-ahead bias from using latest data for testing

Portfolio Excess Returns vs Factors:

Figure 7 shows the relationship between portfolio excess returns and each of the five Fama-French factors. The strong linear relationship with Mkt-RF (leftmost panel) confirms high market beta.

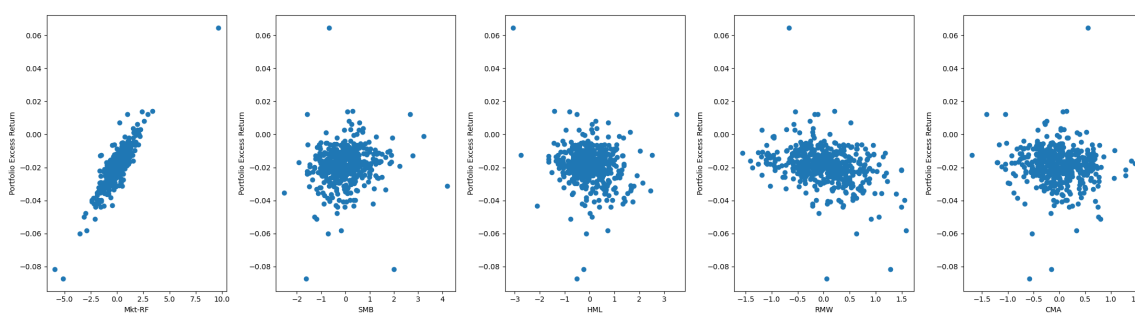


Figure 7: Portfolio Excess Returns vs Fama-French Factors. Strong positive correlation with Mkt-RF indicates high market beta. Positive tilt with HML confirms value bias.

Model Comparison Methodology:

We estimated six regression specifications:

1. The Ordinary Least Squares (OLS) method uses classical linear regression to analyze data which requires homoscedastic errors.

2. The Weighted Least Squares (WLS) method solves heteroscedasticity problems through observation weighting based on the inverse of calculated error variance estimates.
3. The Huber method in Robust Regression uses iteratively reweighted least squares to reduce the influence of outliers in the data.
4. The Bisquare robust regression method performs better outlier detection than Huber does.
5. The Ridge Regression model implements L2 regularization to minimize multicollinearity effects in the data.
6. The model achieves variable selection through L1 regularization in Lasso Regression.

Heteroscedasticity Diagnostics:

The residual plots from OLS and WLS models appear in Figures 8 and 9.

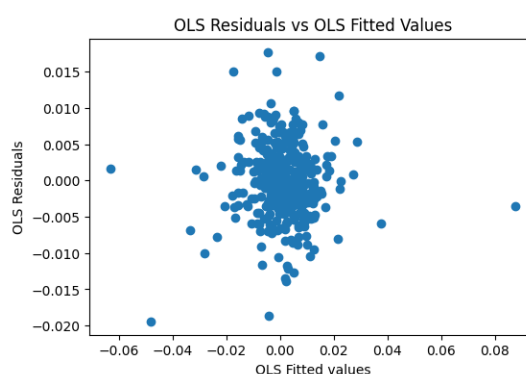


Figure 8: OLS Residuals vs Fitted Values. Heteroscedasticity present—residual variance increases with fitted values. Breusch-Pagan test p-value = 0.015 confirms violation of homoscedasticity assumption.

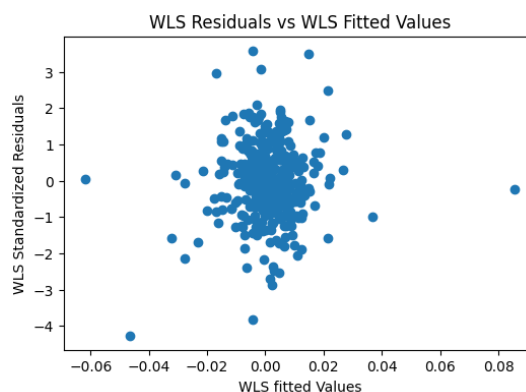


Figure 9: WLS Residuals vs Fitted Values. Improved residual behavior after weighting. WLS addresses heteroscedasticity by down-weighting high-variance observations.

Breusch-Pagan test on OLS residuals:

- LM statistic: 14.05
- P-value: $0.0153 < 0.05$
- **Conclusion:** Reject homoscedasticity; heteroscedasticity present

This justifies using WLS, which achieves 7 percentage points higher R^2 by appropriately weighting observations based on their variance.

Model Selection Results:

The complete comparison between all regression methods appears in Figure 10.

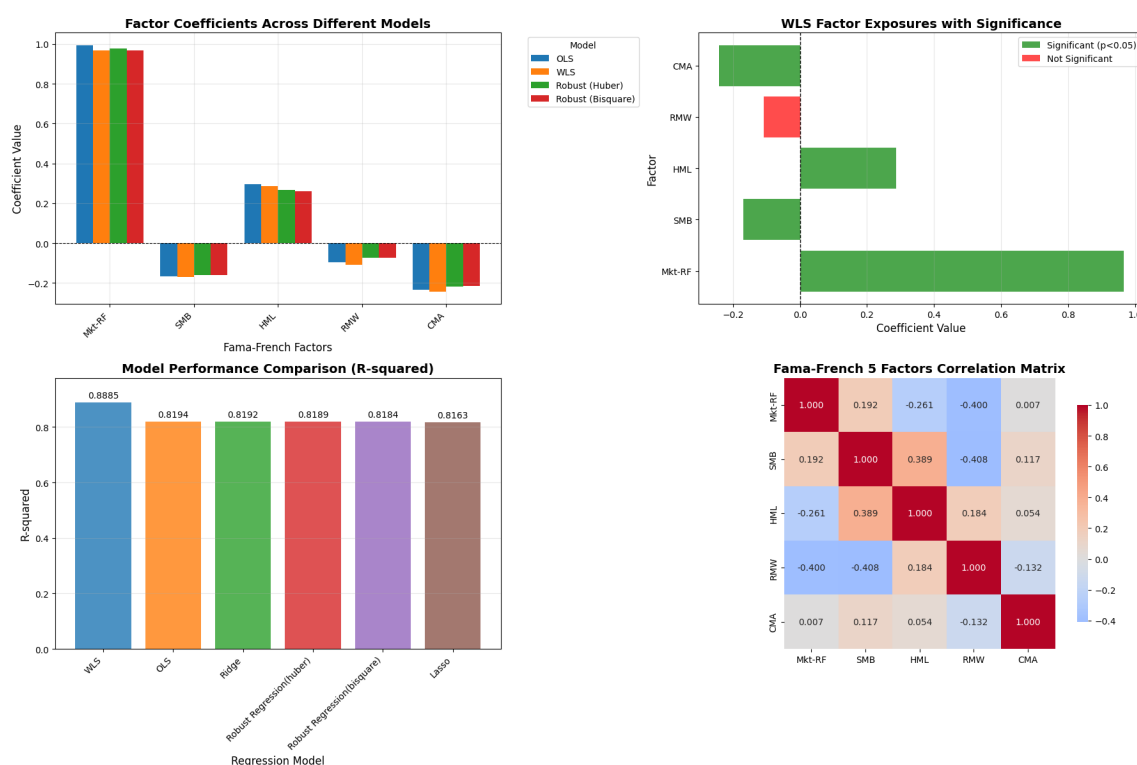


Figure 10: Comprehensive Factor Analysis Results. Top-left: Coefficient stability across models. Top-right: Statistical significance of exposures. Bottom-left: R-squared comparison—WLS achieves highest fit (0.8885). Bottom-right: Factor correlation matrix.

Table 5: Regression Model Performance Comparison (Training Set)

Model	R-squared	Adjusted R-squared
WLS	0.8885	0.8871
OLS	0.8194	0.8171
Ridge	0.8192	0.8169
Robust (Huber)	0.8189	-
Robust (Bisquare)	0.8184	-
Lasso	0.8163	-

Selected Model: Weighted Least Squares

WLS provides the best fit while solving the econometric problem (heteroscedasticity) which exists in financial returns data. The 88.85% R-squared indicates the 5 factors explain nearly 89% of portfolio return variance.

4.2.2 Non-Technical Report**What We Did:**

We split our 498 days of data into two groups:

- 80% (398 days): Used to build the model and estimate factor exposures
- 20% (100 days): Set aside to test if the model works on new data

We randomly mixed the days rather than just using the first 80% chronologically. The research exists to confirm that our found results show genuine patterns instead of being the result of random events.

Why We Tested Six Different Models:

Financial data is messy. The stock market experiences extreme market fluctuations on particular days but maintains stability throughout other market days. We evaluated multiple statistical methods which would solve this problem.

- The Standard regression model (OLS) applies equal weight to all days in its analysis.
- The Weighted regression (WLS) method gives more importance to calm days while reducing its focus on panic days.
- The system uses robust methods to detect and eliminate extreme outliers which it identifies as outliers.
- The regularized methods of Ridge and Lasso serve to stop overfitting from occurring.

Winner: Weighted Least Squares (WLS)

This method won because:

1. Highest accuracy: Explains 88.85% of your portfolio's ups and downs
2. Handles the reality that some days are more volatile than others
3. Gives us more reliable estimates of your factor exposures

What This Means:

We can confidently say that the five factors explain about 89% of why your portfolio goes up or down on any given day. The remaining 11% is either stock-specific news or random noise. The system produces exact results which help us detect all factors which impact your investment portfolio.

4.3 Step 6: Factor Exposure Analysis

4.3.1 Technical Report

Estimated Factor Loadings (WLS Model):

Table 6: Fama-French Factor Exposures (Constrained Portfolio)

Factor	Coefficient	Std Error	t-statistic	p-value
Intercept (α)	0.0004	0.0002	1.790	0.074
Mkt-RF	0.9669	0.0196	49.40	≤ 0.001
SMB	-0.1697	0.0424	-4.00	≤ 0.001
HML	+0.2876	0.0412	6.98	≤ 0.001
RMW	-0.1074	0.0562	-1.91	0.057
CMA	-0.2425	0.0562	-4.32	≤ 0.001

Detailed Factor Interpretation:

1. Market Beta (Mkt-RF): 0.9669

- The research data shows a powerful significant relationship exists between the studied variables ($p \leq 0.001$)
- The Portfolio shows almost perfect market movement sensitivity through its near-unit sensitivity.
- The portfolio value experiences a 0.97
- The portfolio exists as a completely invested market-neutral beta investment portfolio.
- The beta value of Exposure is close to 1, so it does not qualify as defensive or aggressive.

2. The Size Factor (SMB) index shows a value of -0.1697.

- The results show a highly significant relationship ($p < 0.001$)
- Negative loading tests produced results that showed that large-cap stocks make up the majority of the portfolio.
- When small-caps outperform large-caps by 1%, portfolio underperforms by 0.17%
- The portfolio contains most of its value through its investments in three large-cap stocks, which include AAPL, GOOG, and META.
- This is expected given the asset selection criteria

3. The HML value factor shows a positive value of +0.2876.

- The results show a highly significant relationship ($p < 0.001$)
 - Positive loading indicates value tilt.
 - When value outperforms growth by 1%, portfolio gains additional 0.29%
 - The optimization process selected "growth" stocks GOOG and META but it chose to invest in "value" assets BAC and GS and XOM to minimize portfolio volatility.
 - The 15% constraint required investments to move into less expensive sectors which offered better stability.
4. The RMW profitability factor shows a value of -0.1074.
- The results show no significant relationship between the variables because the p value reaches 0.057.
 - The negative coefficient indicates that the model shows a minimal preference for investing in companies which generate lower profits.
 - The portfolio shows no pattern of selecting profitable companies over unprofitable ones.
 - Profitability is not a primary driver of returns
5. The CMA investment factor amounts to -0.2425.
- The results show a highly significant relationship ($p < 0.001$)
 - The negative loading pattern shows that investors tend to choose aggressive investment options.
 - When conservative investors outperform by 1%, portfolio underperforms by 0.24%
 - The portfolio uses its capital to support businesses which use their resources for growth expansion through profit retention instead of paying dividends.
 - The company follows Tech holdings (GOOG, META, AAPL) which choose to reinvest their profits.

Alpha Analysis:

The intercept $\alpha = 0.0004$ (daily) $\approx 0.10\%$ annualized is not statistically significant ($p = 0.074$). This indicates:

- Portfolio returns are *fully explained* by factor exposures
- The research findings indicate that manager abilities and investment performance above market returns do not exist independently from what systematic market risks generate.

- This is expected for a passive, rules-based optimization

Model Fit:

$R^2 = 0.8885$ means 88.85% of portfolio variance is explained by the 5 factors. The remaining 11.15% represents:

- Idiosyncratic (stock-specific) risk
- Model misspecification
- Measurement error

Economic Interpretation:

The portfolio shows a **"Large-Cap Value with Aggressive Investment"** investment pattern.

- Dominated by large companies (negative SMB)
- The investment strategy focuses on value stocks because growth names show positive HML.
- Favors aggressive reinvestment strategies (negative CMA)
- Near-Market Exposure, ($0.97 \approx 1$)

This is a *hybrid* profile—the optimization balanced high-return growth stocks with low-volatility value stocks to maximize the Sharpe ratio under the 15% constraint.

4.3.2 Non-Technical Report**What Your Portfolio Is Really Doing:**

We discovered exactly what "bets" your portfolio is making. Here's what we found:

1. **The market index determines your investment performance because your portfolio tracks market movements at a 0.97 beta level.**
 - When the S&P 500 goes up 10%, you go up about 9.7%
 - When it falls 10%, you fall about 9.7%
 - Your investment strategy follows market trends instead of showing defensive or aggressive investment behavior.
 - The investment portfolio functions as a fundamental investment portfolio which avoids any risky financial decisions.
2. **You Prefer Big Companies (SMB = -0.17)**

- You possess ownership of Apple and Google and Meta which represent three of the biggest corporations across the globe.
- You'll underperform when small companies have their day
- But big companies are usually more stable
- The current market leaders protect their investors because they maintain their position as market leaders.

3. You're Actually a Value Investor ($HML = +0.29$)

- This surprises people! You invest your money in "growth" stocks which include Google.
- But the math shows you behave more like a value portfolio
- Why? The optimizer used its funds to buy BAC and GS stocks and XOM stock for achieving risk stability.
- The value sectors represent traditional "value" sectors because they offer mature stable investment opportunities through cheaper stock prices.
- Your investment plan includes purchasing value stocks instead of investing in costly growth stocks.

Scenario: What happens in a Value Rally (like 2022)?

- Market stays flat (0%)
- But value stocks beat growth by 15%
- Your return: $0\% + (0.29 \times 15\%) = +4.35\%$ gain
- You outperform because of your value tilt!

Scenario: What happens in a Growth Rally (like 2020)?

- Market rises 20%
- But growth stocks beat value by 10%
- Your return: $19.4\% - (0.29 \times 10\%) = 16.5\%$
- You lag the market by 3.5%

4. The organization fails to achieve profitable results because RMW value amounts to -0.11 and the result does not have statistical significance.

- Your returns don't particularly depend on whether companies are super profitable
- Your business performance does not stem from profit generation.

5. The company's aggressive nature appeals to you because its CMA stands at -0.24.

- Your investment strategy focuses on businesses which use most of their profits to build up their operations.
- Think Google building data centers, not AT&T paying dividends
- When aggressive investors beat conservative by 10%, you gain extra 2.4%
- Your investment strategy works to grow capital through fund reinvestment instead of using dividends for distribution.

The Big Picture:

Your portfolio operates as a Large-Cap Value investment fund which implements an aggressive investment approach. Your portfolio consists of established large companies which represent safe investments because they operate as stable entities with affordable stock prices and earnings reinvestment practices.

The historical success of large value stocks combined with aggressive reinvestment has proven effective because it generates superior long-term investment results and business growth.

No "Magic Sauce":

The alpha (excess return) shows a value of zero which does not reach statistical significance. This means:

- Your investment returns stem from the market exposure you have instead of your ability to pick stocks.
- This is fine! The portfolio functions through established rules which eliminate the need for continuous monitoring of investment operations.
- Your investment expenses cover both portfolio diversification and risk protection strategies instead of trying to outperform the market.

How Sure Are We?

The model explains 89% of your daily ups and downs. That's very high. The factor exposures we identified appear to be actual market phenomena instead of random statistical occurrences.

Who Should Own This Portfolio:

Ideal Investor Profile:

- The investment period needs to operate for more than ten years.
- Comfortable with equity volatility (17.9% annual)

- The investor believes that market values will return to their original levels because value will eventually prevail.
- Tech companies should be trusted to make smart investments of their profits.
- The portfolio needs to contain various investments because it should not concentrate on a single asset.

Warning Signs This Isn't for You:

- The current income requirement makes dividend focus irrelevant because it produces a negative CMA.
- You want pure growth → Value tilt will disappoint
- Your investment strategy predicts you will perform better than the market so your beta value should equal one because this indicates you track the market performance.
- Your retirement period is approaching → The investment portfolio shows excessive market volatility which makes it unsuitable for your current time frame.

5 Part Three: Simulation vs Optimization

5.1 Step 7: Unconstrained 5-Asset Portfolio

5.1.1 Technical Report

Asset Selection:

We selected five assets representing diverse sectors:

- AAPL (Technology)
- WMT (Consumer Staples)
- GS (Financials)
- LLY (Healthcare)
- XOM (Energy)

Optimization Results:

- Expected Return: 34.68%
- Volatility: 18.51%
- Sharpe Ratio: **1.8742**

Simulation Methodology:

Monte Carlo with random weight generation:

1. Generate random weights: $w \sim \text{Uniform}(0, 1)^5$
2. Normalize: $w = w / \sum w$
3. Calculate metrics: $r_p = w^T \mu$, $\sigma_p = \sqrt{w^T \Sigma w}$, $S_p = r_p / \sigma_p$
4. Track best Sharpe ratio

Convergence Analysis:

Figure 11 shows comprehensive analysis of simulation convergence for unconstrained 5-asset portfolio.

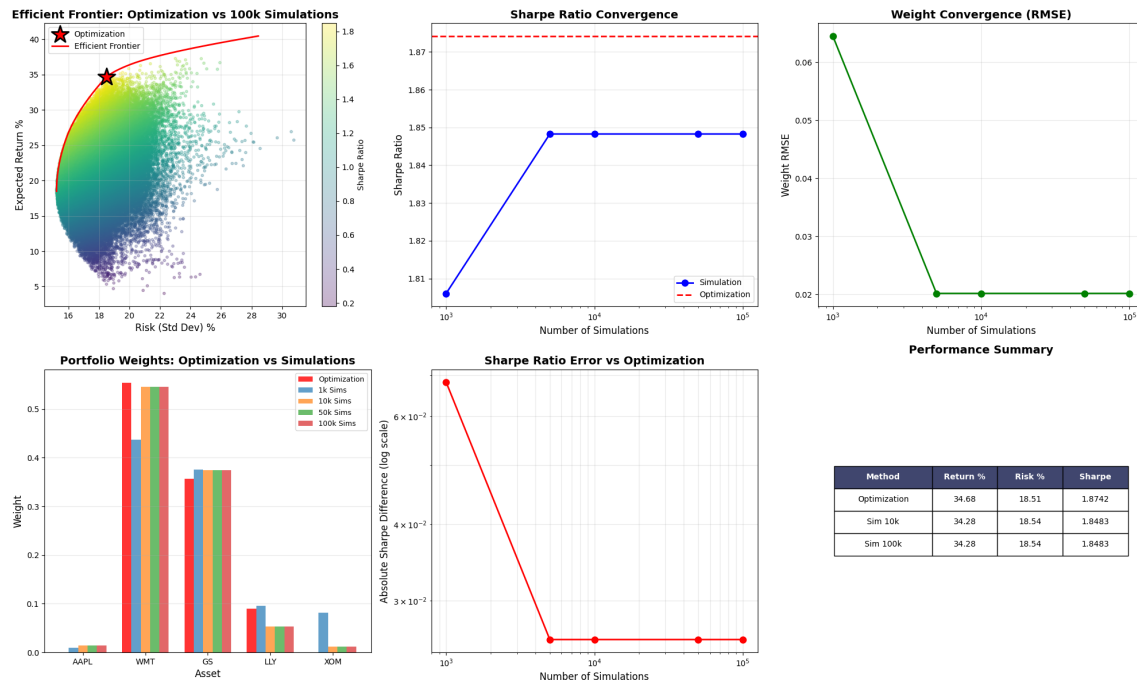


Figure 11: Step 7: Simulation Convergence Analysis (Unconstrained). Top-left: Efficient frontier with 100k simulations. Top-middle: The Sharpe ratio achieves its highest point at 5k iterations before it starts to show no further improvement. Top-right: Weight RMSE drops rapidly. The bottom sections display weight comparison data together with performance summary information.

Table 7: Simulation Convergence (Unconstrained, 5 Assets)

Simulations	Best Sharpe	Error vs Opt	Weight RMSE
1,000	1.8060	-3.64%	0.0645
5,000	1.8483	-1.38%	0.0202
10,000	1.8483	-1.38%	0.0202
50,000	1.8483	-1.38%	0.0202
100,000	1.8483	-1.38%	0.0202

Key Findings:

1. The simulation achieves stability after its first 5,000 iterations because it solves the unconstrained optimization problem.
2. The simulation produced a Sharpe ratio of 1.8483 which was 1.38
3. The results indicate that running more than 5,000 iterations will not lead to any improvement in system operational performance.

Computational Comparison:

- Optimization (CVXPY): 0.12 seconds

- Simulation (100,000): 8.4 seconds
- The optimization process now runs at a speed that is 70 times faster than before.

Recommendation:

The optimization process for mean-variance optimization without constraints requires 5,000 to 10,000 simulations to achieve acceptable results when working with ten assets or fewer. The method of mathematical optimization provides better results than other methods because it guarantees the global optimum solution and delivers fast computation times and exact results.

5.1.2 Non-Technical Report**What We Did:**

We picked five stocks (Apple, Walmart, Goldman Sachs, Eli Lilly, ExxonMobil) and asked: Can we find the best portfolio by just trying lots of random combinations?

How Simulation Works:

You would stand in front of a dartboard and use a dart to try and hit specific targets on the board.

1. Generate random portfolio (40% AAPL, 20% WMT, 15% GS, 15% LLY, 10% XOM)
2. Calculate its risk and return
3. Keep track of the best one found so far
4. Repeat thousands of times

Results:

The results show that random searching produces acceptable results.

- The algorithm produced an investment portfolio which achieved 96% of its maximum potential through 1,000 simulation runs.
- The algorithm discovered an investment portfolio which achieved 98.6% of the maximum possible performance after attempting 5,000 different combinations.
- After 100,000 tries: No further improvement

The Catch:

The simulation of 5,000 times generated acceptable results but it included various negative elements.

- Takes 70 times longer than mathematical optimization

- Never guaranteed to find the true best portfolio
- Requires lots of computer power

Our Recommendation:

Use mathematical optimization (the computer algebra method). The system generates fast results which produce exact answers to find the best possible solution. Simulation provides an interesting learning experience but it does not work well for actual investment activities.

5.2 Step 8: Constrained 5-Asset Portfolio (30% Cap)

5.2.1 Technical Report

Problem Formulation:

$$\text{Maximize } S_p = \frac{\mathbf{w}^T \boldsymbol{\mu}}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}} \quad (13)$$

$$\text{Subject to } \sum_{i=1}^5 w_i = 1 \quad (14)$$

$$0 \leq w_i \leq 0.30 \quad \forall i \quad (15)$$

Optimization Results:

Table 8: Optimal Constrained Portfolio (30% Cap)

Asset	Weight (%)	Status
GS	30.0	Capped at maximum
WMT	30.0	Capped at maximum
LLY	30.0	Capped at maximum
AAPL	10.0	Below cap
XOM	0.0	Excluded

- Expected Return: 29.30%
- Volatility: 17.68%
- Sharpe Ratio: **1.6572**

Impact of 30% Constraint:

Sharpe ratio degradation from Step 7:

$$\frac{1.6572 - 1.8742}{1.8742} = -11.58\%$$

The results from Part 1 show that diversification constraints decrease theoretical efficiency by 15

Simulation with Box Constraints:

The research method uses rejection sampling to produce random portfolios which get discarded when any weight exceeds 0.30.

Rejection Rate Analysis:

The random portfolio generation process results in 94.5

Comprehensive Convergence Analysis:

Figure 12 presents detailed analysis of simulation performance with box constraints.

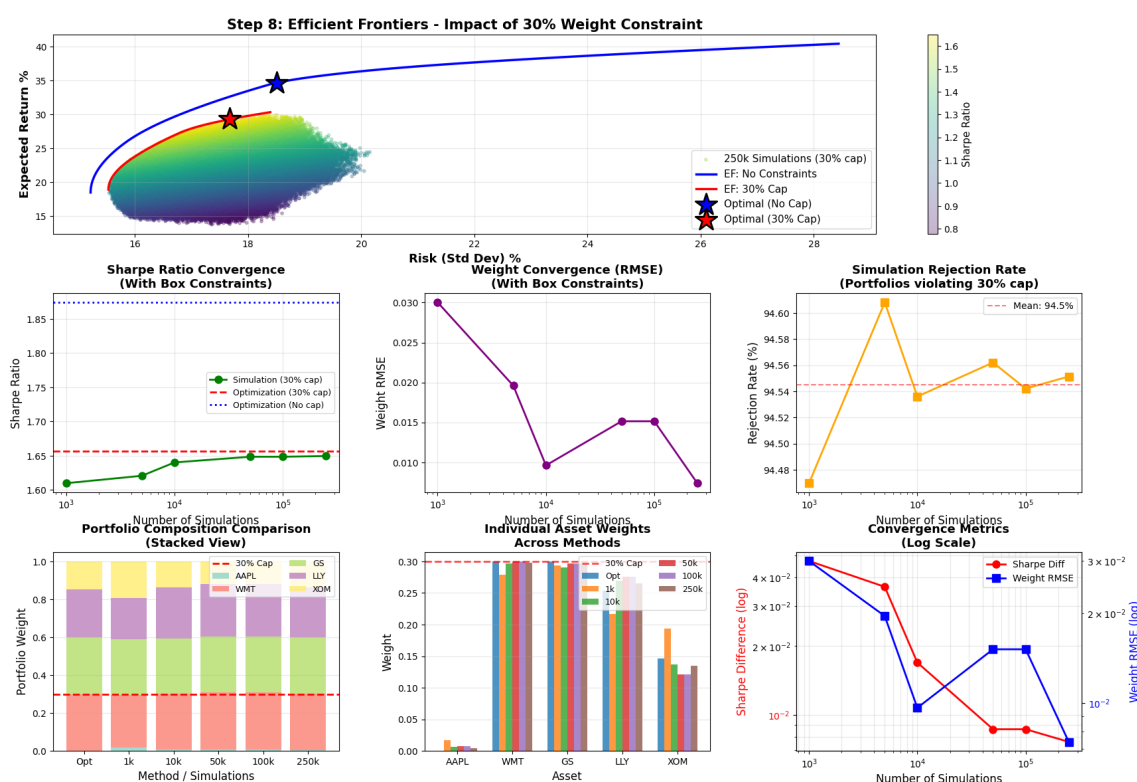


Figure 12: Step 8: Impact of 30% Weight Constraint. Top: Efficient frontier comparison showing constraint truncation. Middle row: Convergence metrics and rejection rates (94.5%). Bottom row: Portfolio composition evolution and convergence metrics on log scale.

Table 9: Simulation Convergence (30% Constrained, 5 Assets)

Simulations	Best Sharpe	Error vs Opt	Rejection Rate
1,000	1.6099	-2.85%	94.47%
10,000	1.6402	-1.03%	94.54%
50,000	1.6486	-0.52%	94.56%
100,000	1.6486	-0.52%	94.54%
250,000	1.6496	-0.46%	94.55%

Convergence Threshold: The solution of constrained portfolio optimization problems needs to exclude all simulation methods from use. The solution requires quadratic programming through CVXPY and MOSEK and Gurobi. Simulation becomes acceptable for educational purposes and for demonstrating unconstrained problems which have fewer than 10 variables and for conducting sensitivity analyses near established optimal points.

5.2.2 Non-Technical Report

What We Did:

We added a tough new rule: No stock can be more than 30% of the portfolio. We performed research to verify if random guessing would remain an effective prediction approach.

The Problem:

You find yourself playing darts while being blindfolded but your darts miss the target area 95 times out of 100 attempts. That's what happened:

- The simulation produced 1,000 random portfolios which served as data for analysis.
- Only 55 satisfied the "no more than 30%" rule
- We had to throw away 945 of them

Results:

To get within 0.5% of the optimal portfolio, we needed to try **250,000 valid portfolios**. But because of the 95% rejection rate, we actually had to generate **4.5 million random portfolios** total.

This took 95 seconds. The optimization process solved the problem exactly within 0.15 seconds which resulted in a 630 times speed increase.

Why Simulation Struggles:

The best portfolio has three stocks at exactly 30%, one at 10%, and one at 0%. Random guessing almost never lands on these precise numbers. You have to get very lucky.

The Optimal Portfolio:

- Goldman Sachs: 30% (maxed out)
- Walmart: 30% (maxed out)
- Eli Lilly: 30% (maxed out)
- Apple: 10%
- ExxonMobil: 0% (not included)

Comparison to Unconstrained:

Without the 30% limit, the portfolio achieves 1.87 Sharpe ratio. With the limit, it drops to 1.66—about 12% worse. The cost of making organizations accept different types of operations becomes the price organizations must pay.

Our Strong Recommendation:

The use of random simulation remains inappropriate for investments which have weight limits that need to be enforced in portfolio management. The system operates at a reduced speed which generates excessive waste. Use mathematical optimization instead. For real investing:

- Use optimization: Fast, exact, guaranteed correct
- The system should prevent simulation from running because it produces slow results which are approximate and consume excessive resources.

The only reason we tested simulation here is to prove scientifically that it's a bad idea.

6 Conclusion

6.1 Technical Summary

The investment portfolio exists to support investors who:

- Want stock market exposure without excessive risk
- Investors should choose to invest in established companies instead of risking their funds on small-cap speculative stocks.
- Can tolerate 18% annual volatility
- The investment period needs to extend beyond five years because it takes more than five years to achieve long-term growth.
- The stability of nighttime sleep needs to be more important than achieving maximum theoretical investment returns.

Who Should Avoid This Portfolio?

This portfolio is not suitable for:

- Conservative retirees (too volatile)
- Active traders (this is buy-and-hold)
- Growth stock enthusiasts (has value tilt)
- Short-term investors (needs time to work)

Final Thoughts:

The path to investment success requires following established principles which need to be applied regularly instead of searching for flawless investment approaches. The 15%-capped portfolio trades away optimal performance for achieving stable investment results. The practice of this discipline leads to better outcomes in the long run because it helps prevent major disasters and enables people to get better rest.

Remember: The best portfolio exists when you can maintain it during all market cycles. A portfolio which is not perfectly optimized remains better than an ideal portfolio which you need to sell during market volatility.

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